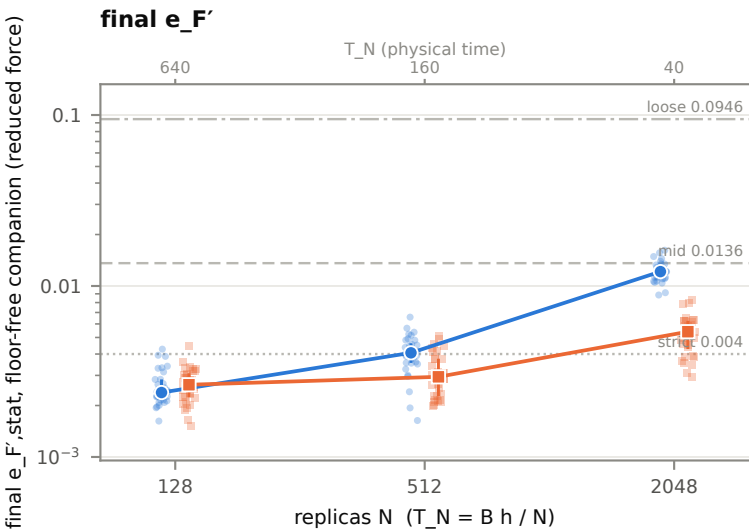
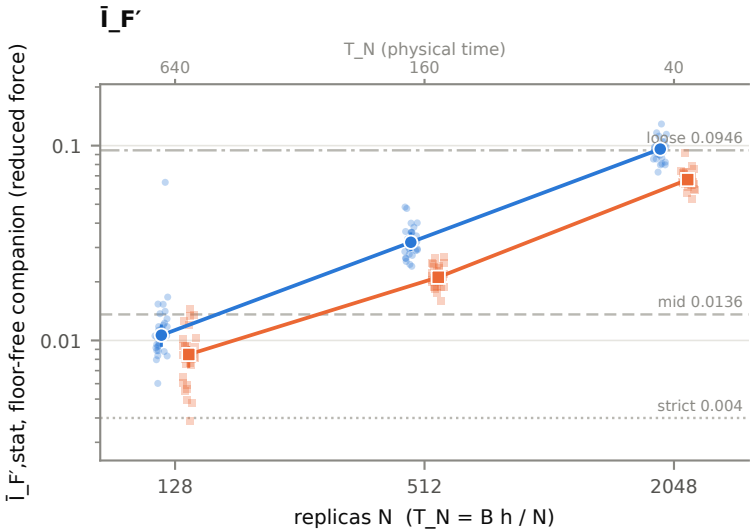
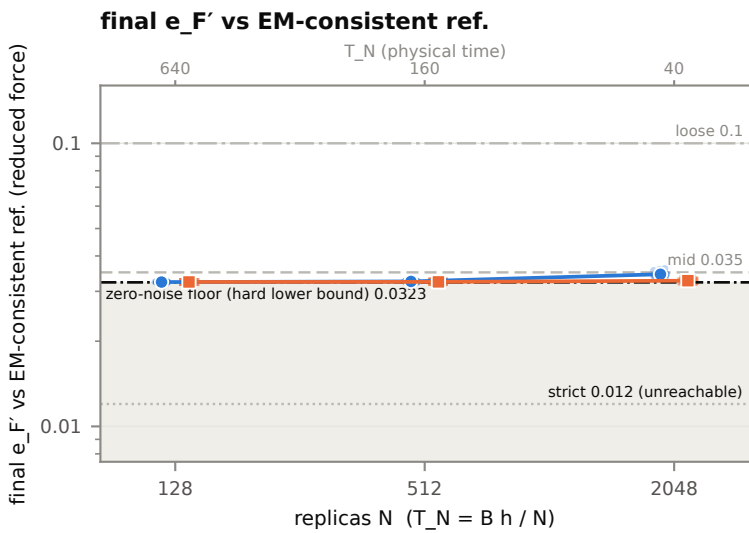
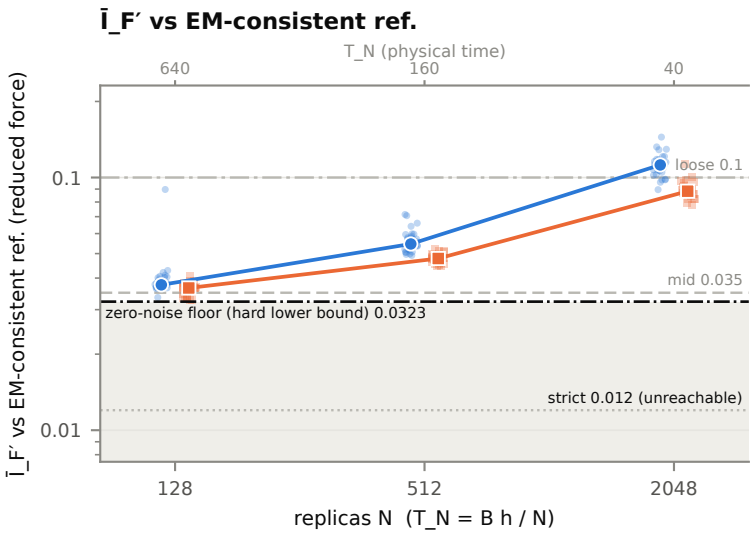
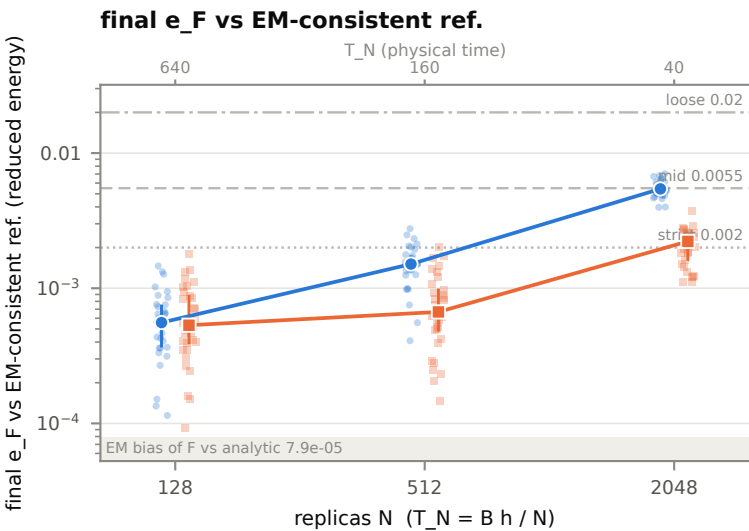
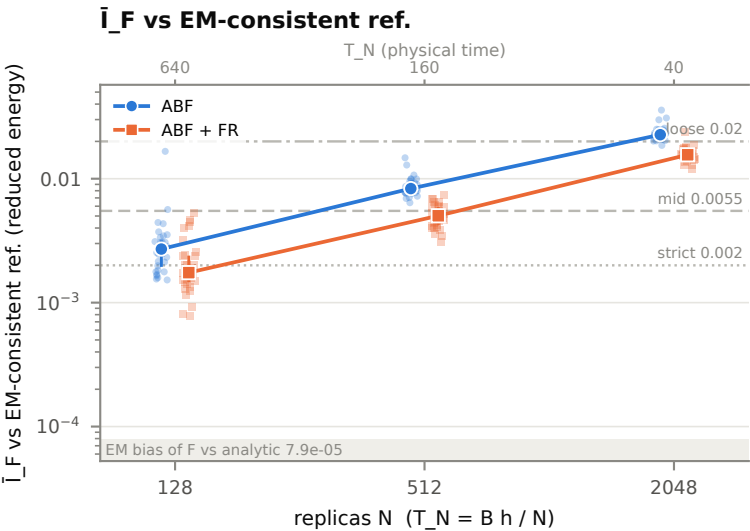


gateway alpha1: secondary errors vs N (EM-consistent reference; floor-free companion of e_F')

Equal total budget per arm per seed: $B = N \times n_steps = 3.2768e+09$ walker-steps at every N ($h = 2.5e-05$, $T_N = B \cdot h / N$); 32 seeds per (N , method). Secondary read-outs reported by the plan (not the primary endpoints). $e_F',stat = \text{RMS over the eval bins of Gamma} - \text{bin-averaged } F'_ref$, so that $e_F'^2 = e_F',stat^2 + \text{floor}^2$ exactly: the companion of e_F' without its deterministic floor (mechanism cell: its own frozen thresholds, the plan's e_F' tau endpoint, are drawn).



gateway alpha1 · S2s_secondary_reference_abs_vs_N · status: all planned (N , seed, method) complete · FR is not defined at $N = 1$ (ABF only, by design). Markers: seed median; whiskers: IQR over seeds; faint points: individual seeds. Grey lines: the frozen strict / mid / loose thresholds of $e(t)$ (shown for scale on the integrated errors as well). Reference floors: $\bar{I}_{Fp_em}$: every estimate lies above the zero-noise floor 0.03227 ($e^2 = stat^2 + floor^2$); thresholds below it are unreachable by construction; $final_e_{Fp_em}$: every estimate lies above the zero-noise floor 0.03227 ($e^2 = stat^2 + floor^2$); thresholds below it are unreachable by construction. Source results/mechanism/matched_free_energy/alpha1/analysis/summary.json (generated 2026-10-10T14:19:18Z, eqb_metrics/2, 10000 bootstrap resamples); plot_synthesis/2.